

Where the attentional bias enters

Two variants of the hypercolumn / point-set model

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Two versions of the basic hypercolumn / point-set model are on the table. They share their architecture completely — a point-set holding a motion hypercolumn and a colour hypercolumn, a single cooperative pool per point-set that is shared across attributes, and divisive normalization that is within attribute and flat across channels. They differ in exactly one respect: where the top-down attentional bias is applied.

That single difference decides whether attending to one feature enhances the other attributes of the same surface by the SAME amount, or by a strictly smaller one. It is therefore the difference the data can see.

Common scaffolding

Index a point-set by p , an attribute by a (motion or colour), and a channel within an attribute by θ — eight directions, eight hues.

$D_{\theta}^a(p, t)$ feedforward stimulus drive to a channel

A_{θ}^a top-down bias: $1 + \beta$ on the attended channel, 1 elsewhere

$E(p, t)$ cooperative pool — ONE per point-set, shared across attributes

$G(p, t) = 1 + \lambda E(p, t)$ the cooperative gain

Both variants then normalize and rectify identically. Writing the numerator of a channel as P :

$$N^a(p, t) = \sum_{p'} w(p, p') \sum_{\theta} P_{\theta}^a(p', t)$$

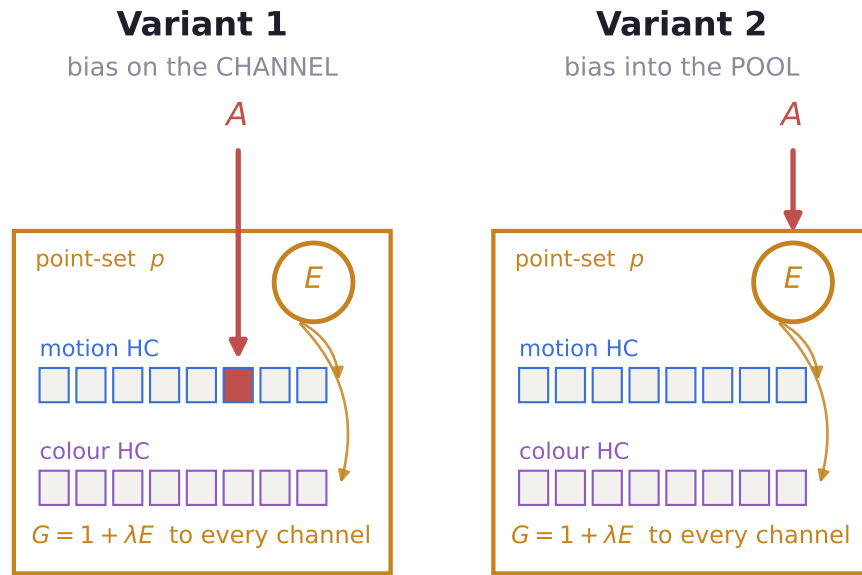
within attribute, across space

$$R_{\theta}^a(p, t) = \frac{[P_{\theta}^a(p, t)]^2}{\sigma^2 + N^a(p, t)}$$

The denominator being FLAT across channels — one per point-set per attribute — is load-bearing. Were normalization per-channel, each channel's denominator would track its own numerator and the shared gain would divide straight back out.

The one structural difference

Everything above is common. The variants differ only in which arrow the bias A travels along:



Variant 1 — bias on the channel

The bias multiplies the stimulus drive of the attended channel directly. That channel is then enhanced twice — once by the bias, and again by the pool gain it helped create — while every other channel in the point-set is enhanced only by the pool gain.

$$P_{\theta}^a = D_{\theta}^a \cdot A_{\theta}^a \cdot G$$

$$\tau_E dE/dt = -E + S\left(\sum_a \sum_{\theta} P_{\theta}^a\right)$$

S = saturating nonlinearity

Variant 2 — bias into the pool

The bias appears only in what drives the pool. It therefore selects which point-sets acquire a strong pool, but once the pool has settled it returns one gain, applied identically to every channel the point-set holds.

$$P_{\theta}^a = D_{\theta}^a \cdot G$$

$$\tau_E dE/dt = -E + S\left(\sum_a \sum_{\theta} A_{\theta}^a D_{\theta}^a\right)$$

What each predicts

Compare a point-set carrying the attended feature (cued, gain G_c) against one that does not (uncued, gain G_u). Because the normalization denominator is flat across channels, the gain does not cancel, and the response ratio for a channel is:

Variant 1

$$\text{attended channel: } R_c/R_u \propto (A G_c/G_u)^2$$

$$\text{every other channel: } R_c/R_u \propto (G_c/G_u)^2$$

Transfer to colour, and to the translation that arrives later, is carried entirely by the pool term. It is real, but strictly smaller than the primary attended feature's enhancement, by the factor A.

Variant 2

$$\text{every channel: } R_c/R_u \propto (G_c/G_u)^2$$

Identical for the attended direction, for colour, and for the translation. Selection is anchored to the point-set, not to the feature.

The measurement that separates them

Take the ratio of the colour index to the primary index within a cued point-set. Variant 2 predicts the two coincide; Variant 1 predicts colour sits below primary by a factor set by beta. No free parameter can move Variant 2 off equality — the equality is structural, which is what makes the comparison worth running.

Open — to settle before either is reported

The two variants are not yet cleanly separated in our own code, and the question is which one we are actually running.

The site's schematic (components/HCPSTFlow.tsx) states that the bias "enters the pool, not the channels, which is why it cannot stay confined to the attribute that was cued". That is Variant 2, explicitly.

But `ps_pointset.py`'s engine reads $PSP = K \times FB(\text{like-to-like MT} \rightarrow V1) \times Coop(1 + CoopL \times E) / \text{Norm}$. A like-to-like feedback term is channel-specific, which puts it in the position of A in Variant 1.

Whether those are the same model described twice, or two different models, has not been checked. It determines which set of equations above describes the results we have.